Xiaohan Gao, Ph.D.

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Xiaohan Gao https://gaoxiaohan.com/

Research Summary

Researcher in AI for chip design, with experience in GPU-accelerated optimization, differentiable FPGA routing, and LLM-based Agentic EDA workflows.

Education

2020-2025 **Ph.D., Peking University**
Ph.D. in Computer Science, Center for Energy-efficient Computing and Applications

2016-2020 **B.S., Peking University**
B.S. in Electronics Engineering and Computer Science, Yuanpei College

Work Experience

2025 July - 2026 July **Postdoctoral Fellow, The University of Texas at Austin**
working with Prof. David Pan and Prof. Yusu Wang
- Developed a GPU-enhanced differentiable FPGA routing framework
- Engaged in a Google project on agentic AI for Chip Design flow

Selected Projects

GPU-Accelerated Differentiable FPGA Routing
- Developed a novel differentiable optimization framework for FPGA routing problem, on GPU platform, which outperforms the SOTA framework on FPGA 2024 contest;
- Working on a multi-agent reinforcement learning framework that scales up to tens of thousands of agents for combinatorial problem solving like routing.
[FCCM'26] *DiffRouter: A Differentiable Routing Framework for UltraScale FPGAs*

Agentic Flow for Chip Design
- Developed an agentic workflow of chip design tools for analog circuit automation involving both topology and geometrical information;
- Mentored a small group to implement agentic flow for chip design tools.
[ISCAS'25, TCAD'25] *LayoutCopilot: An LLM-Powered Multiagent Collaborative Framework for Interactive Analog Layout Design*

Optimization Algorithms for Physical Design
- Developed a software based on analytical algorithms for physical design.
[ICCAD'24] *Joint Placement Optimization for Hierarchical Analog/Mixed-Signal Circuits*
[DATE'23] *SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility*
[DAC'21, TCAD'23] *Interactive Analog Layout Editing with Instant Placement and Routing Legalization*

Selected Awards

2025 **Research Forum Best Poster** ASP-DAC
Asia and South Pacific Design Automation Conference 2025

2023 **Best Paper Award** DATE
Design, Automation and Test in Europe Conference 2023

Professional Service

Committee Member

2026 **Technical Program Committee** ISEDA

Program Committee Member, AI4EDA Workshop

2024 **Artifact Evaluation Committee Member** MLCAD Symposium

Review Service

TCAD **Reviewer** *IEEE Transactions on Computer-Aided Design of Integrated Circuits & Systems*

TODAES **Reviewer** *ACM Transactions on Design Automation of Electronic Systems*

Selected Publications

1. **X. Gao**, Z. Xiong, Y. Wang, and D. Z. Pan, "DiffRouter: A Differentiable Routing Framework for UltraScale FPGAs," in *2026 IEEE 34th Annual International Symposium on Field-Programmable Custom Computing Machines*, ser. FCCM'26, 2026, pp. 124–132. DOI: 10.1109/FCCM68464.2026.00029

2. B. Liu, H. Zhang, **X. Gao**, X. Tang, et al., "LayoutCopilot: LLM-Empowered Analog Layout Design towards Enhanced Human-Machine Interaction," in *2025 IEEE International Symposium on Circuits and Systems*, ser. ISCAS'25, 2025, pp. 1–5.

3. **X. Gao**, H. Zhang, B. Liu, Y. Lin, et al., "Joint Placement Optimization for Hierarchical Analog/Mixed-Signal Circuits," in *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design*, ser. ICCAD'24, New York, NY, USA, 2025.

4. H. Zhang, J. Song, **X. Gao**, X. Tang, et al., "EasyACIM: An End-to-End Automated Analog CIM with Synthesizable Architecture and Agile Design Space Exploration," in *2024 The Chips to Systems Conference*, ser. DAC'24, 2024.

5. H. Zhang, **X. Gao**, Z. Shen, J. Song, et al., "SAGERoute 2.0: Hierarchical Analog and Mixed Signal Routing Considering Versatile Routing Scenarios," in *2024 Design, Automation & Test in Europe Conference & Exhibition*, ser. DATE'24, Valencia, Spain, 2024.

6. H. Zhang*, **X. Gao***, H. Luo, J. Song, et al., "SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility," in *Design, Automation & Test in Europe Conference & Exhibition*, ser. DATE'23, Antwerp, Belgium, 2023 **(Best Paper Award)**.

7. Y. Chen, J. Mai, **X. Gao**, M. Zhang, and Y. Lin, "MacroRank: Ranking Macro Placement Solutions Leveraging Translation Equivariancy," in *Proceedings of the 28th Asia and South Pacific Design Automation Conference*, ser. ASPDAC'23, Tokyo, Japan, 2023.

8. Q. Wang, **X. Gao**, Y. Lin, R. Wang, and R. Huang, "DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography," in *Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design*, ser. ICCAD'22, San Diego, CA, USA, 2022.

9. **X. Gao**, M. Liu, D. Z. Pan, and Y. Lin, "Interactive Analog Layout Editing with Instant Placement Legalization," in *2021 58th ACM/IEEE Design Automation Conference*, ser. DAC'21, San Francisco, CA, USA, 2021.

10. **X. Gao**, C. Deng, M. Liu, Z. Zhang, et al., "Layout Symmetry Annotation for Analog Circuits with Graph Neural Networks," in *Proceedings of the 26th Asia and South Pacific Design Automation Conference*, ser. ASPDAC'21, Tokyo, Japan, 2021.

11. **X. Gao***, H. Zhang*, M. Liu, L. Shen, et al., "Interactive Analog Layout Editing with Instant Placement and Routing Legalization," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 42, no. 3, pp. 698–711, 2023.